

TEAM ID	SWTID-2026-4030
PROJECT TITLE	Plugging into the Future: An Exploration of Electricity Consumption Patterns
MAX MARKS	3 MARKS

3.4 – Technology Stack

1. Frontend (User Interface)

- HTML5 – Structure of web pages.
- CSS3 – Styling and responsive design.
- JavaScript – Interactive features.
- Bootstrap – Responsive UI components.
- Chart.js or D3.js – Data visualization and electricity consumption graphs.

2. Backend (Application Layer)

- **Python** – Core programming language for data analysis.
- **Flask** or **Django** – Web framework for handling requests and serving results.
- **REST API** – Communication between frontend and backend.

3. Database

- **MySQL** – Storage of electricity consumption records.
 - Alternative: **PostgreSQL**.
- **Tables:**
 - User Information.
 - Electricity Usage Data.
 - Analysis Results.
 - Forecast Data.

4. Data Processing & Analytics

- **Pandas** – Data cleaning and manipulation.
- **NumPy** – Numerical computations.
- **Matplotlib / Seaborn** – Statistical visualization.
- **Scikit-learn** – Machine learning and forecasting models.

5. Forecasting & Prediction

- Linear Regression.
- Random Forest Regression.
- Time Series Analysis (ARIMA/Prophet).

Used to:

- Predict future electricity demand.
- Identify peak consumption periods.
- Analyse seasonal usage trends.

6. Data Sources

- Smart Meter Data.
- Utility Company Datasets.
- Public Energy Consumption Datasets.
- CSV/Excel Files for historical records.

7. Development Tools

- **Visual Studio Code (VS Code)** – Code editor.
- **Jupyter Notebook** – Data analysis and model development.
- **Git** – Version control.
- **GitHub** – Source code repository.

8. Deployment

- **Apache or Nginx** Web Server.
- **Docker** (optional).
- Cloud Platforms:
 - [Amazon Web Services \(AWS\)](#)
 - [Microsoft Azure](#)